Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

A semester VII Information Technology major project redefining how financial data is stored, shared, and secured — entirely offline, entirely private, entirely yours.

[Student Name] · [Enrollment Number] · Semester VII · Information Technology · [College Name]

The Problem with How Finance Works Today

Every transaction you make leaves a trace — stored on servers you don't control, processed by systems you can't audit, and exposed to breaches you never see coming.

Privacy at Risk

Centralized databases store sensitive financial data in single points of failure, vulnerable to mass breaches and unauthorized access.

Always-Online Dependency

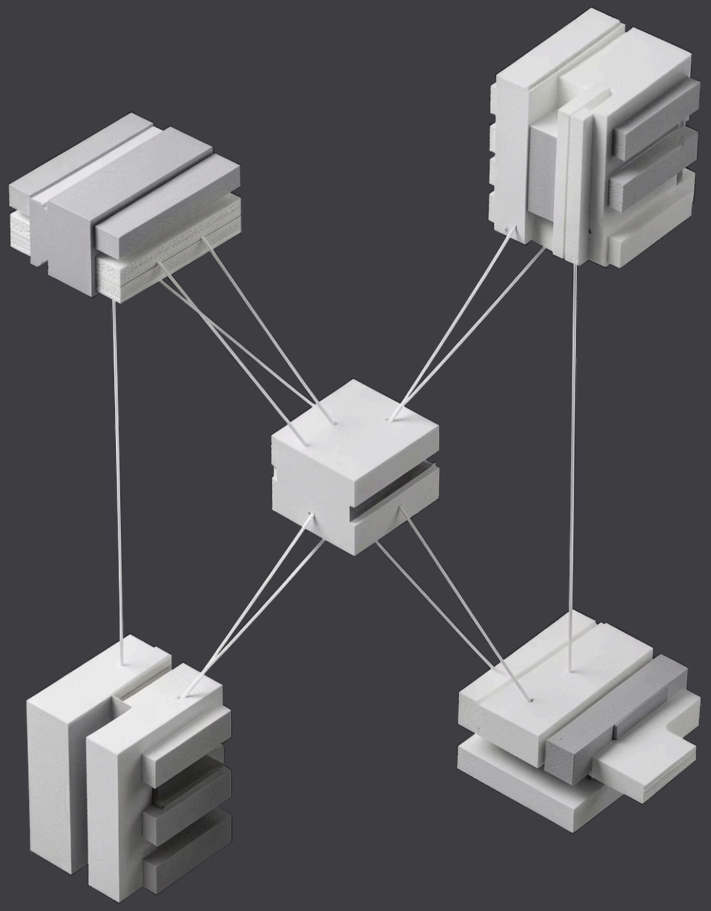
Modern finance collapses without internet connectivity — no payments, no records, no operations in low-connectivity environments.

Recurring Infrastructure Costs

Monthly POS subscriptions, cloud hosting fees, and server maintenance create unsustainable overhead for small merchants.

Zero Data Ownership

Users surrender complete control of their financial history to third-party platforms with opaque data policies.



Introducing Spendly

Three applications. One unified ecosystem. Zero servers. Every component operates independently, communicates directly, and stores nothing in the cloud.

Spendly

The consumer wallet. Track expenses, claim offers, scan barcodes, and manage your complete financial life — offline-first, encrypted at rest.

Spendly Shop

The merchant terminal. Accept payments, generate invoices, manage inventory, and issue QR receipts without any monthly subscription or internet connection.

Spendly Radar

The discovery layer. Broadcast localized offers and promotions peer-to-peer to nearby users — no central server, no tracking, no intermediaries.

System Architecture

Spendly's architecture is built like a precision instrument — each layer has a singular purpose, and every component communicates through well-defined, secure interfaces.

01

Frontend Layer

React PWA with IndexedDB via Dexie.js — fully offline-capable UI with persistent local storage and instant load times.

02

Networking Layer

WebRTC and PeerJS enable direct device-to-device communication. Cloudflare Workers with KV provide optional edge relay for discovery.

03

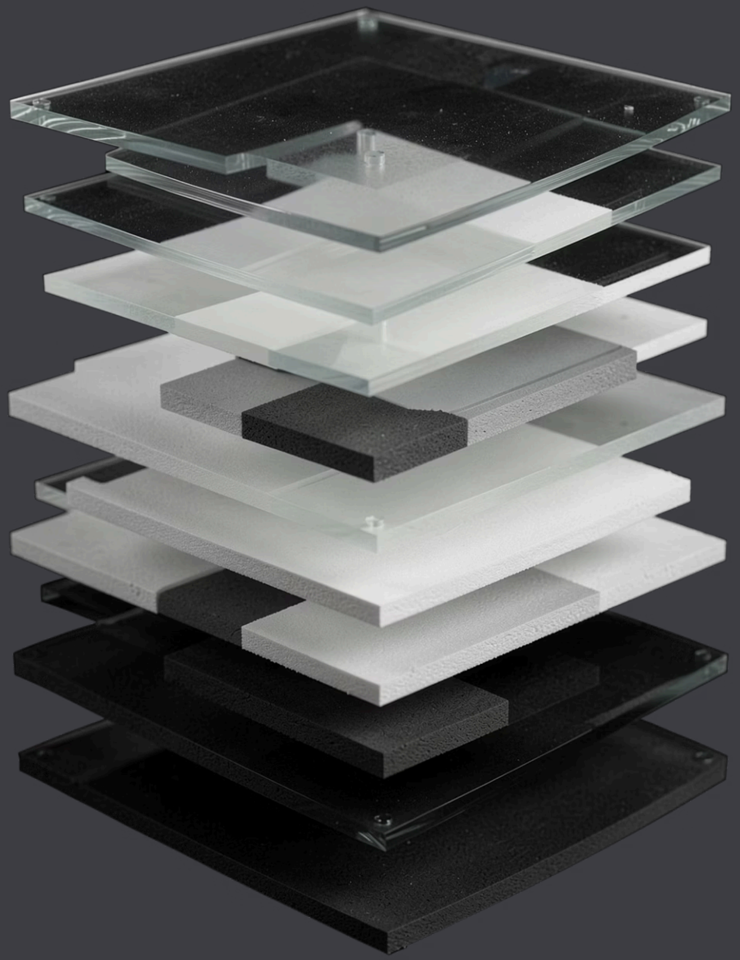
Security Layer

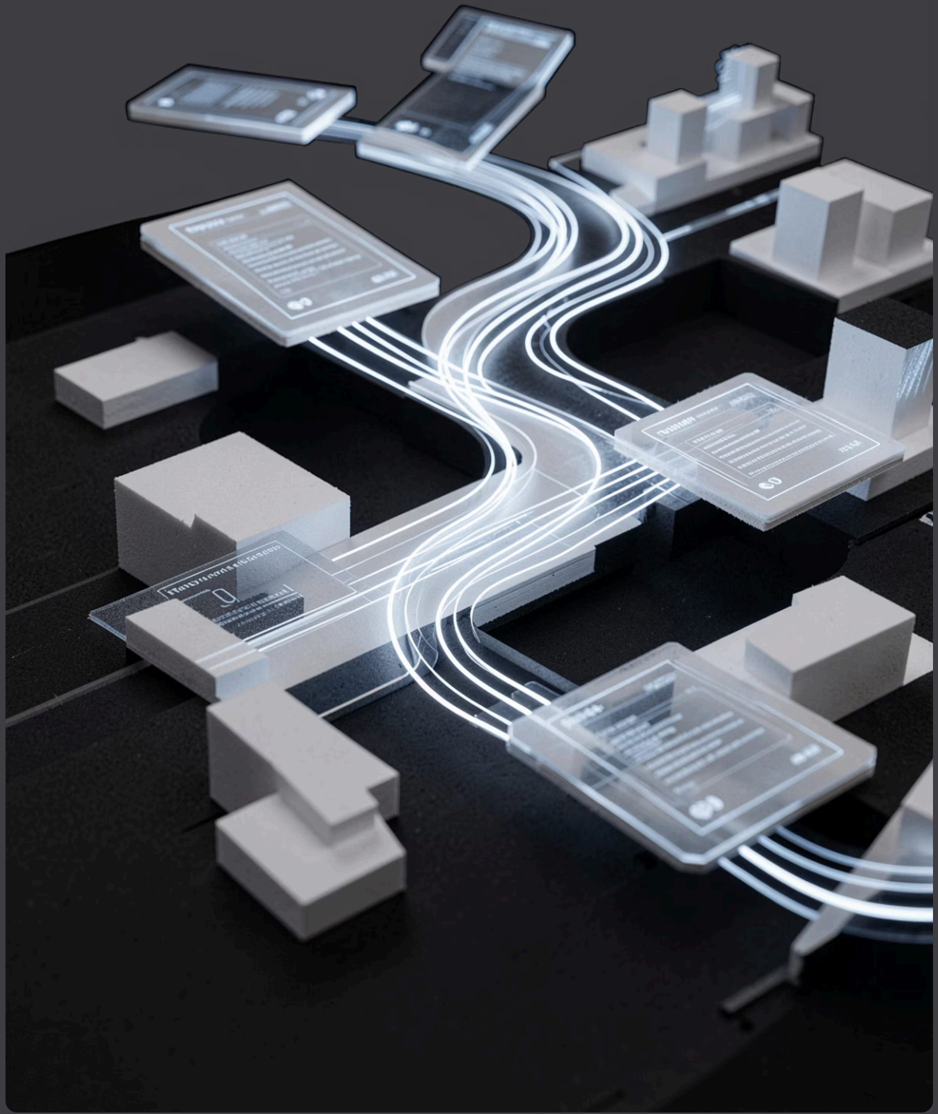
WebAuthn for passwordless authentication. AES-256-GCM encryption for all stored data. Secure Enclave integration for biometric unlock.

04

Deployment Layer

Cloudflare Workers serverless edge functions. Zero persistent infrastructure. Sub-millisecond global response times when online.





How a Transaction Flows

From discovery to settlement, every step happens locally and peer-to-peer. No round-trips to a server. No waiting. No exposure.

1

Discover

User discovers a local offer via Spendly Radar's peer-to-peer broadcast in proximity.

2

Claim

Coupon claimed and stored in encrypted IndexedDB. Barcode generated for in-store redemption.

3

Validate

Merchant scans barcode. Spendly Shop validates offer locally — no server call required.

4

Invoice

Invoice generated, shared peer-to-peer, and archived. Expense recorded. Analytics updated instantly.

Technology Stack

Every technology choice was deliberate. Each tool was selected for its ability to operate without a server, protect data without compromise, and scale without infrastructure.

Layer	Technology	Purpose
Frontend	React + PWA	Installable, offline-first user interface
Database	IndexedDB + Dexie.js	Encrypted local persistence on device
Networking	WebRTC + PeerJS	Direct peer-to-peer device communication
Edge	Cloudflare Workers + KV	Serverless relay and ephemeral state
Security	WebAuthn + AES-256-GCM	Biometric auth and military-grade encryption

Security by Design

Spendly doesn't add security as a feature — security is the foundation. Every architectural decision begins with the question: *how do we protect this without a server?*

Zero-Knowledge Architecture

No financial data ever leaves the device in plaintext. The server knows nothing about your transactions, balance, or identity.

WebAuthn Biometric Unlock

Passwordless authentication via fingerprint or face recognition. Credentials stored in the device's Secure Enclave — never on a server.

AES-256-GCM Encryption

All data at rest is encrypted using authenticated encryption. Tamper detection built in. Keys derived from biometric hardware bindings.

Offline-First Data Model

IndexedDB stores all records locally. Sync is optional and peer-to-peer. The application is fully functional without any network connection.

Core Features

Every feature was designed to replace an existing tool — wallet, POS terminal, expense tracker, loyalty card, and analytics dashboard — in a single, private, offline-capable application.

Expense & Wallet

Real-time expense tracking with a built-in cash wallet. Categorize, filter, and analyze spending — all stored locally.

Merchant Billing & QR Invoice

Generate and share itemized invoices via QR code. No printer required. No paper. No server.

Inventory & EMI Alerts

Track stock levels in Spendly Shop. Set automated EMI and payment reminders for customers and merchants alike.

Offline Sync & Analytics

Full functionality without internet. When connectivity returns, peer-to-peer sync resolves conflicts silently and securely.

Advantages & What Comes Next

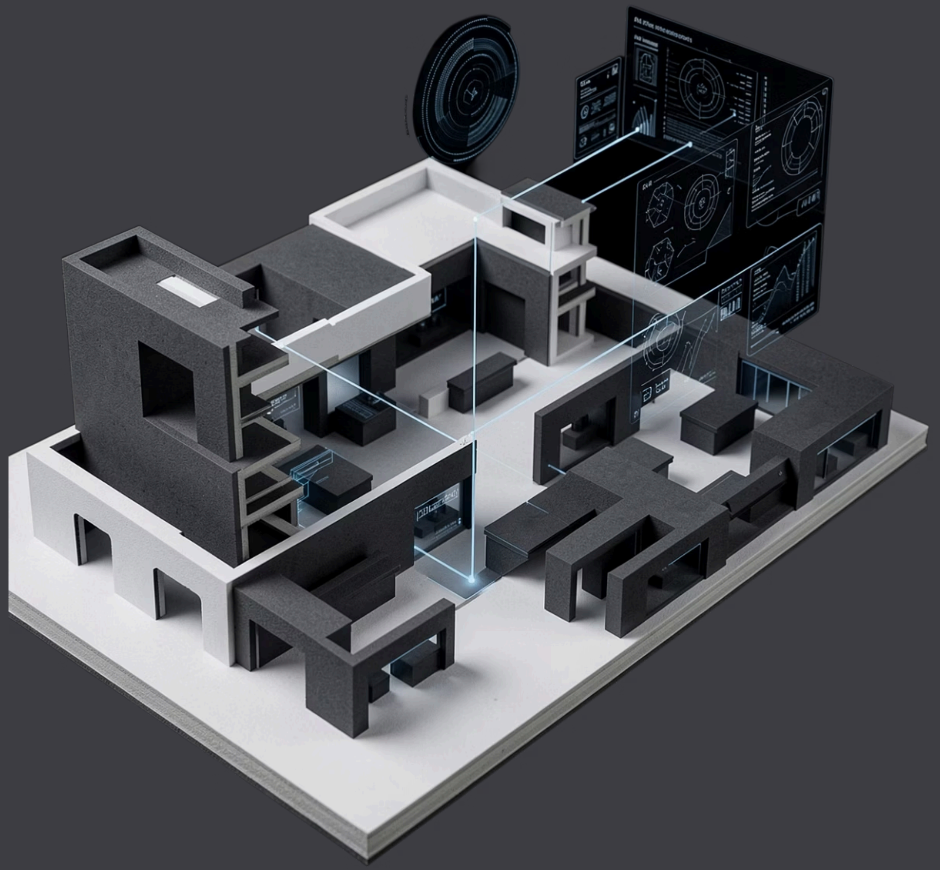
Spendly eliminates the cost, latency, and risk of traditional finance infrastructure — and the roadmap extends that vision further.

Current Advantages

- **Offline First** — Fully operational without internet connectivity
- **Zero Server Cost** — No hosting, no subscriptions, no infrastructure
- **Complete Privacy** — Zero-knowledge architecture, no data leakage
- **Fast Billing** — Instant peer-to-peer invoice generation and sharing
- **P2P Communication** — Direct device-to-device, no intermediaries

Future Scope

- **AI Insights** — On-device spending pattern analysis and predictions
- **Voice Billing** — Hands-free merchant transactions via voice commands
- **UPI & NFC** — Native integration with India's payment infrastructure
- **GST Compliance** — Automated tax invoice generation for merchants
- **Multi-Store** — Unified dashboard for merchant chains and franchises



Spendly

Finance. Rebuilt for privacy. Built for offline. Built for you.

Spendly is not an incremental improvement on existing finance apps. It is a fundamental rethinking of what financial infrastructure should look like when privacy is the default, offline is the norm, and the user owns their data completely.

Offline-First

Works everywhere, always

Zero-Knowledge

No one sees your data

Peer-to-Peer

No servers, no middlemen

Thank you. Questions are welcome.